

RESEARCH ARTICLE

Cognitive reappraisal emotion regulation interventions in the workplace and their impact on job performance: An ecological momentary intervention approach

Ze Zhu¹  | John A. Aitken²  | JeongJin Kim²  | Julia I. Baines²  | Seth A. Kaplan²  | Reeshad S. Dalal²  | Jordan Hassani²

¹Department of Psychological and Brain Sciences, Texas A&M University, College Station, Texas, USA

²Department of Psychology, George Mason University, Fairfax, Virginia, USA

Correspondence

Ze Zhu, 4235 TAMU, College Station, TX 77843, USA.

Email: mia.zhu.io@gmail.com

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Abstract

Emotion regulation at work is important; however, effective emotion regulation is difficult, necessitating timely intervention. We developed an ecological momentary emotion regulation intervention that incorporated two cognitive reappraisal strategies. Using a randomized controlled trial design, we assessed the impact of the cognitive reappraisal intervention on employees' workplace affective experiences and, in turn, counterproductive work behaviour (CWB) and overall job performance. Participants in the between-person intervention and control groups ($N = 88$ and $N = 88$, respectively) completed two cognitive reappraisal or control activities daily for three workweeks and reported on their daily work outcomes. Results revealed that the cognitive reappraisal intervention significantly mitigated negative affect and enhanced positive affect, which in turn reduced CWB and improved overall job performance, respectively. Furthermore, a follow-up 1 month after the end of the intervention revealed no "fade out" of the intervention effect. Finally, among the two cognitive reappraisal strategies tested, results revealed that reappraising the situation was more effective than reappraising the emotion. Substantively, the study provides valuable evidence linking reappraisal-based emotion regulation interventions to sustained improvements in job performance. Methodologically, the

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study provides a *causal* yet *in situ* demonstration of the effectiveness of workplace ecological momentary interventions.

KEYWORDS

cognitive reappraisal, counterproductive work behaviour, ecological momentary intervention, emotion regulation, job performance

Practitioner points

- The cognitive reappraisal ecological momentary intervention effectively mitigates negative emotions and enhances positive emotions, which in turn reduce counterproductive work behaviour and improve overall job performance, respectively.
- The cognitive reappraisal ecological momentary intervention does not exhibit any 'fade out' effect on counterproductive work behaviour or overall job performance 1 month after the intervention ends.
- Organizations can provide employees with ecological momentary emotion regulation interventions to help employees manage their own emotional and behavioral reactions in their typical work environments.
- A comparison of the two cognitive reappraisal strategies shows that reappraising the situation is more effective than reappraising the emotion. Therefore, focusing on reappraising the situation can maximize the effect of the cognitive reappraisal intervention.

INTRODUCTION

Workplace affect—that is, mood and emotions—is associated with critical organizational outcomes such as job performance, turnover and job satisfaction (Barsade & Gibson, 2007; Seo et al., 2004). Short-lived affective experiences reflect individuals' immediate responses to environmental cues and play a crucial role in driving employees' momentary work behaviours (Beal et al., 2005; Weiss & Beal, 2005). Regulating such affective experiences at work is important, yet often challenging (e.g. Gross, 2002; Shockley et al., 2012). Emotion regulation refers to the processes by which individuals experience, perceive, express and attempt to modulate their affective states (Gross, 1998a). Failure to regulate one's affect at work can lead to various harmful effects for the employees themselves as well as their organizations (Dalal et al., 2009; Kaplan et al., 2009).

Solutions that aim to enhance emotion regulation skills place the responsibility on the employees who are already impacted to remember the emotion regulation knowledge obtained via the training, recognize their present affective states, understand the specifics of the situation causing these affective states, determine the right time to apply an emotional regulation strategy and choose the appropriate strategy to use (Lazarus & Folkman, 1984). In contrast, we propose that attempts to help employees manage their affect in their real work environments should account for and incorporate the situationally bound, dynamic and momentary nature of affective states. Accordingly, we suggest an alternative type of intervention for emotion regulation in the workplace: ecological momentary interventions. Ecological momentary *interventions*, which we suggest are the logical next step from ecological momentary *assessments* (also called experience sampling methods), involve delivering interventions in one's natural environments in near-real time, allowing for quick application to the ongoing affective context (Heron & Smyth, 2010; Shim et al., 2021). Unlike standard interventions, ecological momentary interventions in the workplace optimize intervention effectiveness by emphasizing temporal states of vulnerability to

negative effects (e.g. high levels of negative affect at particular times or in particular situations) and/or opportunity for positive effects (e.g. periods of increased susceptibility to positive affect) as well as temporal states of receptivity to interventions (e.g. moments when employees are able to receive, process and utilize the provided intervention support; Nahum-Shani et al., 2015).

To date, as noted by Zhu et al. (2024), ecological momentary interventions have primarily been used in behavioural health research (e.g. smoking cessation; Eghdami et al., 2023) rather than in organizational research. The states of vulnerability that are prime targets for ecological momentary interventions, however, exist not just in health domains but also in the workplace. For example, just as a health researcher could intervene on occasions when a person with a history of problematic drinking is in a vulnerable psychological state that may lead them to seek out alcohol, an organizational researcher could intervene on occasions when an employee is in a particularly negative affective state and therefore vulnerable to enacting counterproductive work behaviour (CWB) or, more generally, to experiencing a dip in overall job performance (Dalal et al., 2009). Therefore, we believe that ecological momentary interventions have considerable potential in organizational research.

Following from these conclusions, we utilize the emotion regulation literature (Gross, 1998a, 1998b) to design an emotion regulation ecological momentary intervention, drawing on Weiss and Cropanzano's (1996) Affective Events Theory to delineate the subsequent job performance outcomes of this intervention. Based on the emotion regulation literature, cognitive reappraisal stands out as one of the most beneficial approaches for emotion regulation (McRae, 2016; Webb et al., 2012). Accordingly, we select two cognitive reappraisal strategies—changing judgements of events (i.e. reappraising the situation) and changing affective reactions to events (i.e. reappraising the emotion)—to target different appraisal processes. Specifically, we assess the potential of a workplace ecological momentary intervention employing and comparing these two cognitive reappraisal emotion regulation strategies to mitigate negative affect (NA) and enhance positive affect (PA)—and, in turn, reduce CWB (Spector & Fox, 2002) and enhance overall job performance. Moreover, we capture the state of receptivity to intervention (i.e. availability and perceived usefulness of the intervention) at that particular time to investigate how receptivity might influence the intervention's impact on outcomes.

To examine the effectiveness of the ecological momentary intervention, we designed a between-person, randomized controlled trial study with an ecological momentary intervention group and a control group. Also, to capture within-person fluctuations in affective and performance outcomes, we conducted an experience sampling study for 3 weeks with both between-person groups, where we administered three daily surveys each workday. Specifically, the first two daily surveys included the intervention or control activities and captured affective outcomes, while the third daily survey assessed job performance outcomes. As described in more detail subsequently, this design allows for a test of the impact of the cognitive reappraisal intervention on affective states, CWB, and overall job performance at the between-person level (by comparing participants in the between-person intervention vs. control groups) as well as comparing the immediate impact on affective states of the two cognitive reappraisal strategies (reappraising the situation vs. reappraising the emotion) against each other and against a control activity at the within-person level (and specifically within only the between-person intervention group, in which participants completed predominantly reappraisal activities but also, occasionally, control activities).

This study makes several contributions. First, this study contributes to an understanding of workplace emotion regulation interventions (Grandey & Melloy, 2017; Troth et al., 2018). Our approach contrasts with the more typical intervention of providing emotion management training once in *ex situ* settings, divorced from the momentary states that necessitate the use of those strategies and the proximal context in which they are used. While existing within-person intervention studies (Hülshéger et al., 2015; Lee & Madera, 2021; Ng & Diener, 2013; Song et al., 2018) offer valuable insights into the effects of cognitive reappraisal, these studies typically deliver interventions at either the start or the end of the workday, thus failing to capture near-real-time affect and events. Additionally, one study (Hülshéger et al., 2015) only utilizes self-report measures of intervention usage, rather than implementing the intervention itself. Our study, conversely, enhances these interventions by incorporating momentary states

and proximal contexts during the workday. We do so by assessing employee affect immediately before and after the intervention and by concurrently measuring employee receptivity to the intervention.

Second, our study aims to examine the active ingredient of the intervention—cognitive reappraisal—rather than evaluating a broader intervention program (e.g. mindfulness interventions that incorporate a blend of cognitive reappraisal, attentional deployment and expressive suppression strategies; Gross, 2015). This approach allows us to better understand the underlying mechanisms driving the effectiveness of the intervention.

Third, this study compares the effectiveness of two established cognitive reappraisal strategies (Webb et al., 2012) against each other. The process model of emotion regulation (Gross, 1998a) suggests that reappraising the situation and reappraising the emotion both belong to the cognitive change phase. However, these two cognitive reappraisal strategies target different aspects of affective stimuli (i.e. external vs. internal aspects of affective stimuli; Uusberg et al., 2019). It is unclear which reappraisal technique is more effective. As one of the first studies to compare the two cognitive reappraisal strategies within a single study (vs. across different studies with potentially different foci, research designs and types of participants) in the workplace, this research makes a novel contribution by not only distinguishing them conceptually but also testing their effectiveness *in situ*.

Fourth, our study examines both the proximal affective and distal performance outcomes of cognitive reappraisal. Notably, we provide the first evidence of the impact of a cognitive reappraisal ecological momentary intervention on daily CWB and overall job performance. Moreover, by examining CWB and overall job performance not just daily during the intervention program but also 1 month after the end of the intervention program, this study tackles a common concern in intervention research across a wide variety of domains (e.g. Arbel et al., 2024; Reizer & Mey-Raz, 2019; van Aar et al., 2017): namely, whether the effect of the intervention ‘fades out’ soon after the intervention is no longer administered. ‘Fade out’ is arguably especially important to examine in the context of ecological momentary interventions because such interventions involve near-real-time support and focus on momentary changes (i.e. proximal outcomes)—and it is an open question as to whether they lead to the acquisition of lasting skills and long-term benefits (Lee & Madera, 2021).

Finally, an important methodological contribution of this study involves providing *causal* tests of theory-based predictions using an ecological momentary intervention design. The current design mitigates concerns about reverse causality and omitted causal variables that bedevil regular ecological momentary assessment designs (Beal, 2015). The current design also aligns the levels of analysis and timeframes of the research design with existing affect theories that emphasize within-person dynamics (e.g. Kuppens et al., 2010; Weiss & Cropanzano, 1996). Moreover, the design merges the precision of experimental lab studies in establishing causal affect–behaviour links with the practicality of field research by implementing a rigorous experimental design in real-world contexts.

HYPOTHESIS DEVELOPMENT

In this section, we develop our theoretical arguments and articulate study hypotheses and research questions. The overall conceptual model can be seen in Figure 1.

The effect of cognitive reappraisal intervention on negative and positive affect

Cognitive reappraisal involves reshaping the understanding of one or more aspects of affectively charged stimuli, thereby modifying its affective impact (Gross & John, 2003; Troy et al., 2023). Reappraisal theories suggest that cognitive reappraisal can enhance affective outcomes through reconstruing either a situation or an affective response to a situation (Uusberg et al., 2019, 2023). For example, individuals can adopt alternative mental models to interpret an event or repurpose a situation or affective response

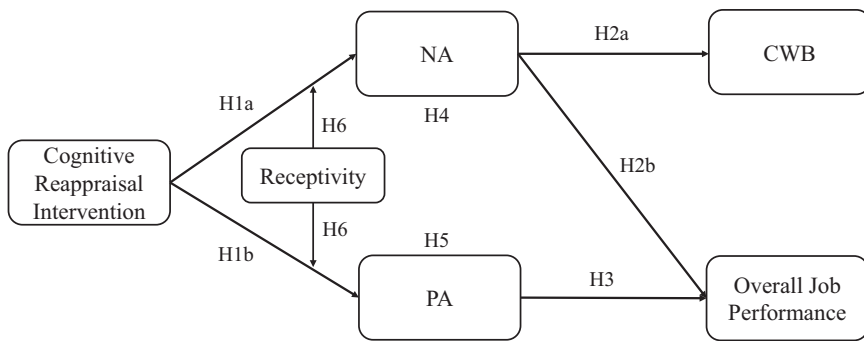


FIGURE 1 Conceptual model. Hypothesis 4 is the mediation hypothesis involving the indirect effects of the cognitive reappraisal intervention on CWB and overall job performance via NA (H4a and H4b, respectively). Hypothesis 5 is the mediation hypothesis involving the indirect effect of the cognitive reappraisal intervention on overall job performance via PA. CWB, counterproductive work behaviour; H, hypothesis; NA, negative affect; PA, positive affect.

by changing some aspects of their ongoing goals to better align with the current situation or affective states, respectively (Uusberg et al., 2023). We focus on broad affective constructs—negative affect and positive affect (NA and PA, respectively)—rather than discrete emotions as the proximal outcomes of intervention for the following reasons. First, research has shown that cognitive reappraisal interventions can have effects on many different discrete emotions (e.g. anxiety: Feldner et al., 2006; amusement: Giuliani et al., 2008). Because the instructions of cognitive reappraisal interventions are typically not designed to target a specific emotion (Webb et al., 2012), we chose to capture NA and PA to align with the design of the intervention (McRae et al., 2012). Moreover, at a higher level of abstraction, discrete emotions have commonly been categorized into positive and negative affect (Ashkanasy & Dorris, 2017; Briner & Kiefer, 2005). Thus, measuring NA and PA is a more efficient approach than measuring multiple discrete emotions. Third, from a practical perspective, given that employees can experience a wide variety of discrete emotions at work (Ashkanasy & Dorris, 2017; Lazarus et al., 2001), if interventions are to be practically applicable in organizational settings, it is helpful for them to target broad affective states rather than one or two discrete emotions.

In a *reappraising the situation intervention*, individuals are instructed to reassess or adjust their interpretation of an affect-eliciting situation so as to highlight its possible positive or meaningful aspects (Denny, 2020; Wang et al., 2021). There are two categories of mechanisms through which such an intervention can change affective outcomes.

First, reappraising the situation can involve the reinterpretation of a negative situation. Specifically, one underlying mechanism for reappraising the situation is that it works against the human tendency to pay disproportionate attention to negative events (Baumeister et al., 2001; Bono et al., 2013). In addition, a negative situation may initially be interpreted as threatening (Lazarus & Folkman, 1984), and reappraising the situation can alleviate the perceived threat by distinguishing non-threat-predictive stimuli from threat-predictive stimuli (Bono et al., 2013; Riepenhausen et al., 2022). Thus, reappraising the situation helps down-regulate NA by altering negative semantic and perceptual representations of the situation.

Second, paying attention to positive aspects of the situation can be viewed as a cognitive adaptation that allows individuals to have access to positive memories and incorporate positive meanings into their interpretation of the world and themselves (Bono et al., 2013; Janoff-Bulman, 1992). Research suggests that reappraisal enables individuals to reconstrue challenging situations as opportunities for growth, learning and meaning-making (Folkman & Moskowitz, 2000; Park & Folkman, 1997). This cognitive process can not only diminish the initial negative interpretation but also construct new positive meanings, allowing individuals to derive benefits from unfavourable circumstances. For instance, an employee facing a critical project setback who initially views it as a

career-threatening failure might reappraise the situation, instead seeing it as a good opportunity to develop problem-solving skills and demonstrate leadership in crisis management. Furthermore, reappraising the situation in a positive way can activate the resources associated with the positive aspects of the situation, thereby reducing the perceived threat of insufficient coping resources for a situation (Lazarus & Folkman, 1984). Thus, reappraising the situation may also improve the up-regulation of PA through the recognition of potential benefits or meaning in the situation (Quoidbach et al., 2015; Riepenhausen et al., 2022).

In contrast, in a *reappraising the emotion intervention*, individuals are asked to view their internal affective reactions to an event as normal, healthy and temporary reactions and, therefore, to openly accept any experienced affect as genuine yet also, importantly, transitory (Kuehner et al., 2009; Troy et al., 2018). While reappraising the emotion and mindfulness interventions both emphasize the importance of being attentive to ongoing experiences in an open and non-judgmental way (Baer, 2003), a mindfulness intervention is a broader composite approach that blends cognitive reappraisal with other categories from the process model of emotion regulation (Gross, 2015), including attentional deployment and expressive suppression.

Accepting one's affective states in a non-judgmental way removes the need for active regulation (Dunn et al., 2009), thus efficiently achieving desired affective outcomes (e.g. reduced NA and/or increased PA; Wenzel et al., 2020). One underlying mechanism for reappraising the emotion is the non-reactivity to inner affective experience: allowing feelings to come and go without becoming fixated on them, so that individuals are not stuck in the habitual reactions to their external and internal environments (Hölzel et al., 2011). In other words, reappraising the emotion improves affective responses by overriding the automatic response to *judge* affective states (Bargh et al., 1992). Indeed, research shows that accepting affective experiences in a nonjudgmental way is associated with the self-perception of decreased reactivity (Carmody & Baer, 2008).

We propose that reappraising the emotion can ameliorate negative affective experiences and enhance positive ones. In our intervention, participants are instructed to openly accept NA without labeling them as 'bad' or 'unwanted'. This non-reactive stance allows individuals to pause and notice the disturbing affective experiences without reacting, potentially weakening habitual maladaptive affective responses and reducing NA (Hölzel et al., 2011). Although reappraising the emotion primarily aims to reduce NA, it may also enhance PA by fostering a mindful state that improves overall affective experience (Glomb et al., 2011; Good et al., 2016; Sawyer et al., 2022). Furthermore, consistent with affect literature suggesting that NA and PA are interdependent at the within-person level (Weiss & Cropanzano, 1996), reducing NA may concurrently promote PA, positioning the intervention as effective for both down-regulating NA and up-regulating PA.

Several correlational and experimental studies support the effectiveness of these two types of cognitive reappraisal strategies in reducing NA and promoting PA (Boemo et al., 2022; Troy et al., 2023). Moreover, human neuroimaging studies have demonstrated that reappraisal engages frontal and cingulate regions implicated in cognitive control to regulate affective responses in systems such as the amygdala and insula (Buhle et al., 2014; Morawetz et al., 2017; Ochsner & Gross, 2008). However, these studies predominantly involve laboratory settings, undergraduate students and/or induced NA, limiting the ability to generalize to work settings involving employee populations and naturally occurring NA (and PA). As mentioned earlier, even the few within-person interventions in the work setting (Lee & Madera, 2021; Song et al., 2018) did not actually deliver an intervention during the workday to capture momentary affective states. Therefore, extending the previous research, we propose that a cognitive reappraisal ecological momentary intervention can have similar effects on employees' affective states at work. Specifically, we propose the following hypothesis:

Hypothesis 1. The cognitive reappraisal intervention leads to: (a) lower post-intervention state NA and (b) higher post-intervention state PA, after controlling for pre-intervention state NA and PA, respectively.

Post-intervention affect and work behaviours

Having previously outlined the theoretical and empirical foundation for how cognitive reappraisal can effectively regulate emotions, we now describe affect–performance links. It should be noted that with regard to affect–performance links, we focus on post-intervention affect rather than on the pre- to post-intervention change in affect. This is because post-intervention affect aligns more closely with our theoretical model and the literature on state affect and job performance (Beal et al., 2005; Weiss & Cropanzano, 1996), which emphasizes affect *levels* (vs. *changes* in levels) as predictors of within-person performance outcomes. In this regard, we describe mechanisms linking NA and PA to CWB and overall job performance, respectively.

First, NA should impact CWB. According to Affective Events Theory (Weiss & Cropanzano, 1996) and the stressor-emotion model of CWB (Spector & Fox, 2005), as well as empirical tests thereof (e.g., Dalal et al., 2009; Sawhney & McCord, 2023), NA stemming from negative work events and job stressors is a leading cause of CWB. The NA–CWB link has been theorized to occur due to two pathways (Dalal et al., 2009; Weiss & Cropanzano, 1996). First, employees experiencing NA may engage in CWB as a way to cope with NA (Matta et al., 2014; Penney & Spector, 2005) by actively addressing problematic work situations through retaliation-type actions (e.g. aggression) or by escaping the negative situations through avoidance-type actions (e.g. withdrawal behaviour; Chen & Spector, 1992; Dalal et al., 2009). Such retaliation- and avoidance-type behaviours, as manifestations of CWB, may repair NA by reciprocating the negative social exchange (Matta et al., 2014). Second, NA increases the accessibility of negative judgements, such as negative views of human nature and unfavourable opinions of others (Barsky & Kaplan, 2007). CWB, in turn, is the behavioural manifestation of these negative judgements (Dalal, 2005; Fox et al., 2001). For instance, when an employee is in a bad mood, they may perceive interactions with colleagues to be more conflict-laden and consequently may hide information from colleagues or enact other forms of CWB towards them (Penney & Spector, 2005). Turning to PA, given that previous research has both conceptually suggested (Spector & Fox, 2005) and empirically demonstrated (e.g. Dalal et al., 2009; Sawhney & McCord, 2023), a weak relationship between PA and CWB, we do not propose a formal hypothesis regarding the PA–CWB relationship.

With regard to affect and overall job performance, research has consistently shown that NA can impair employee performance (George & Zhou, 2007; Kaplan et al., 2009; Shockley et al., 2012). According to affective events theory (Weiss & Cropanzano, 1996), NA can hinder employee performance due to the diversion of cognitive resources from primary tasks to emotion management, leading to rumination over affective events and difficulty concentrating on work-related tasks (see also Merlo et al., 2018; Weiss & Merlo, 2020). This is further exacerbated by NA narrowing attentional focus, making responses in dynamic work environments less flexible (Fredrickson, 1998, 2001). On the other hand, PA enhances job performance by broadening cognitive resources and improving attentional control (Beal et al., 2005; Fredrickson, 2001, 2013). PA also boosts work motivation, leading to the selection of challenging goals and increased persistence (George & Brief, 1996; Isen, 2001; Tice et al., 2007).

In addition to having an impact on performance at the daily level, the effect of the intervention should persist beyond the conclusion of the intervention. The timely delivery of cognitive reappraisal intervention should enhance individuals' affective awareness, equip them with skills to critically appraise the impact of affective events on their affective experiences, and empower them with strategies to manage their affective challenges in a timely manner, all of which can be transferred to future emotion regulation occasions (Kay & Skarlicki, 2020; Yeung & Wong, 2020). Thus, participants who receive the cognitive reappraisal ecological momentary intervention are likely to employ more adaptive emotion regulation strategies in future affective situations and, thus, to exhibit less CWB and better overall job performance even after the end of the intervention period. In other words, we do not expect the effects of the cognitive reappraisal intervention to rapidly 'fade out' once interventions are no longer being delivered (Kam et al., 2024). Given that the underlying theoretical rationale is the same, we will not make a distinction, in Hypothesis 2 and subsequent hypotheses, between performance outcomes measured at

the daily level during the intervention period and those measured 1 month after the conclusion of the intervention period. Rather, encompassing both timeframes, we more generally propose:

Hypothesis 2. Post-intervention state NA is: (a) positively related to CWB and (b) negatively related to overall job performance.

Hypothesis 3. Post-intervention state PA is positively related to overall job performance.

Impact of the intervention on distal outcomes via affect

A cognitive reappraisal intervention is designed to shift how employees interpret affective situations or the resultant affective states, directly reducing NA and enhancing PA (Gross & John, 2003), which in turn lowers CWB and improves overall job performance (Spector & Fox, 2005; Weiss & Cropanzano, 1996). In essence, our cognitive reappraisal intervention, as an emotion regulation ecological momentary intervention, is expected to improve affective outcomes that serve as proximal outcomes (i.e. the immediate goals of the intervention; Zhu et al., 2024) to the intervention. These proximal outcomes are in turn expected to further impact distal outcomes (i.e. the ultimate goals of the intervention; Zhu et al., 2024), such as CWB and overall job performance, in multiple ways.

First, CWB often results from negative affective experiences that are poorly managed (Spector & Fox, 2005). Cognitive reappraisal can help employees navigate negative affective experiences more effectively (i.e. reduce NA), thus reducing the likelihood of engaging in CWB (Gross, 2002). For instance, an additional work project, originally interpreted as more work, could be reinterpreted as a learning opportunity, thus improving affect and, in turn, reducing CWB. Moreover, such interventions could help employees accept NA as natural and transitory, thus mitigating the strength of these reactions and, in turn, reducing CWB.

Similarly, the indirect effect should extend to overall job performance. By helping to facilitate adaptive responses to stressful and challenging situations (Gross & John, 2003), cognitive reappraisal can reduce NA and increase PA, which, in turn, promotes overall job performance (Beal et al., 2005). For example, reduced NA and increased PA can broaden cognitive resources and improve attentional control for work-related tasks, leading to better problem-solving capabilities, concentration and persistence (Troy et al., 2013; Weiss & Merlo, 2020), which are important for job performance. Hence, we hypothesize

Hypothesis 4. Post-intervention state NA mediates the relationships between the cognitive reappraisal intervention and: (a) CWB and (b) overall job performance.

Hypothesis 5. Post-intervention state PA mediates the relationships between the cognitive reappraisal intervention and overall job performance.

Receptivity as a moderator

Receptivity has been highlighted as a crucial aspect of ecological momentary interventions, as employees are not equally receptive to interventions on all occasions (Nahum-Shani et al., 2018). Receptivity refers to the moments when individuals are open to receiving, processing and utilizing the provided intervention support (Nahum-Shani et al., 2015). It reflects the recipients' ability and/or motivation to utilize a specific intervention on any given occasion (Zhu et al., 2024). To capture both the ability and motivation conceptualizations of receptivity, we focus on two separate indicators of receptivity on the part of the recipient: (a) the employee's availability to engage in the intervention on that occasion and (b) the perceived usefulness of the intervention to the employee on

that occasion. If individuals are less receptive for either reason, they should be less likely to invest effort in the intervention activity (Nahum-Shani et al., 2023). This would result in a diminished effect of the intervention on outcomes (Nahum-Shani et al., 2022, 2023). Thus, we explore whether employee availability and the perceived usefulness of the intervention moderate the relationship between the cognitive reappraisal intervention and the proximal outcomes of post-intervention state NA and PA. Specifically, we propose that stronger relationships between the intervention and both reduced NA and increased PA will occur when employee availability and/or the perceived usefulness of the intervention to employees are high, as opposed to low.

Hypothesis 6a. Employee availability strengthens the relationship between the cognitive reappraisal intervention and post-intervention state NA and PA, after controlling for pre-intervention state NA and PA, respectively.

Hypothesis 6b. The perceived usefulness of the intervention to the employee strengthens the relationship between the cognitive reappraisal intervention and post-intervention state NA and PA, after controlling for pre-intervention state NA and PA, respectively.

Comparing the two types of cognitive reappraisal intervention

In this study, we focus on two variations within the cognitive reappraisal strategy, which show theoretical similarities and differences. Both reappraisal strategies—reappraising the situation and reappraising the emotion—fit within the cognitive change stage of the process model of emotion regulation (Gross, 1998a). This stage emphasizes altering the meaning or interpretation of one or more aspects of the affective experience before it sustainably influences affective outcomes (Gross & John, 2003). Additionally, both strategies may operate through similar underlying psychological mechanisms. Research suggests that reconstrual (reframing a stimulus to alter its meaning) and repurposing (finding new meaning or purpose in an affective response) are core processes that support the effectiveness of cognitive reappraisal strategies (Uusberg et al., 2019, 2023). These shared mechanisms reflect how both strategies facilitate affective outcomes by changing one's cognitive interpretation of experiences.

Despite the aforementioned similarities, however, the two cognitive reappraisal strategies also differ in important ways. The primary difference lies in the target of reappraisal. Reappraising the situation focuses on external stimuli, modifying one's interpretation of external events that cause the affective responses (Ellsworth & Scherer, 2003). In contrast, reappraising the emotion targets internal stimuli, modifying one's internal affective responses to the external stimuli (Uusberg et al., 2019). Another distinction lies in the direction of the reappraisal. Reappraising the situation often involves identifying positive aspects—for example, reappraising a setback as a learning opportunity. In contrast, reappraising the emotion adopts a more neutral stance: individuals reframe their affective experiences as neutral reactions, thereby helping to reduce the need for active regulation (Dunn et al., 2009). Given the aforementioned differences, the two strategies may offer distinct pathways for emotion regulation, potentially leading to different affective outcomes. While initial meta-analytic evidence has shown no significant differences between the two strategies (Webb et al., 2012), the primary studies in the meta-analysis were predominantly conducted at the between-person level in laboratory settings with undergraduate participants. The intervention and context of the current study—an ecological momentary intervention in a workplace setting—are different from the previous studies, and thus there is no clear evidence about whether one cognitive reappraisal intervention is more effective than the other in this context. Thus, we compare the two strategies via a research question:

Research Question 1: Is there a difference in the impacts of the two cognitive reappraisal interventions on post-intervention state NA and PA, after controlling for pre-intervention NA and PA, respectively?

Of note, although (as indicated previously) our ultimate focus is on the between-person effects of the intervention, our investigation of this research question, which focuses on comparing the two cognitive reappraisal interventions to each other (and to the control activity), instead occurs at the within-person level, and indeed only for the subset of participants who were assigned to the between-person intervention (vs. control) group. Specifically, we randomly assigned participants *within* the between-person intervention group to receive one of the two reappraisal activities on 90% of occasions and a neutral control activity on the remaining 10% of occasions. This design resembles that of a within-person micro-randomized trial, which is designed to allow for causal tests of the immediate effect of an intervention activity at the within-person level of analysis (Klasnja et al., 2015), within the between-person intervention group of our overall randomized controlled trial design. Such a research design, in which the between-person intervention group actually includes more than one intervention as well as (on a small number of occasions) a control activity, also follows best practices in the ecological momentary intervention literature aimed at reducing participant boredom with the intervention (Nahum-Shani et al., 2018). Thus, our test of this research question allows us to determine whether the intervention's effects are attributable to occasions on which participants (in the between-person intervention group) have been randomly assigned to complete the reappraising the situation activity, the reappraising the emotion activity or the control activity.

METHOD

Sample

We conducted an *a priori* power analysis to test the between-person main effect of the intervention with the following specifications: alpha at .05, power at .80, a specified correlation between repeated measures at .50 (i.e. the default setting without any different *a priori* expectations), and a conservative effect size of .20 (i.e. between a small and a medium effect size, but one-third closer to the latter, according to Cohen, 1992). The minimum required sample size was 150 participants (75 per group).

We recruited participants through online advertisements, physical flyers and snowball sampling. To be eligible for the study, individuals needed to (1) be at least 18 years old, (2) work 30 or more hours a week, (3) have a day-shift work schedule from Monday to Friday and, in order to facilitate payment, (4) be a US citizen or permanent resident. Of the 183 participants who completed the initial survey, we retained data from those who had responded to at least three daily surveys (see Gabriel et al., 2018; Puranik et al., 2021), resulting in a final sample size of 176 participants.¹ Of this final sample, 43 employees were recruited from a non-profit organization, 18 from non-academic employees at a large mid-Atlantic university, 32 from an online experience sampling participant panel maintained by one of the authors of this paper, 27 from an online advertisement posted on Craigslist and 56 via snowball sampling.

The average age of participants was 33.93 years ($SD = 7.90$), 57.95% of participants identified as female, and 73.86% identified as White (11.93% identified as Black or African American and 10.80% as Asian). The modal (30.68%) job tenure was 2–5 years, and 43.75% of participants worked in organizations that had 500 or more employees. Most of the sample were hybrid workers (80.11%) and 17.91% of the sample were supervisors (vs. individual contributors). Participants in the sample worked an average of 40.47 h/week ($SD = 3.59$). The industries wherein the most participants worked were professional, scientific or technical services (21%), manufacturing (20%), educational services (15%), and health care or social assistance (9%). This study was approved by the university's Internal Review Board.

¹We included the requirement of completing at least three daily surveys due to the statistical requirement that a person needs to provide at least three within-person observations to facilitate within-person analysis (Gabriel et al., 2018).

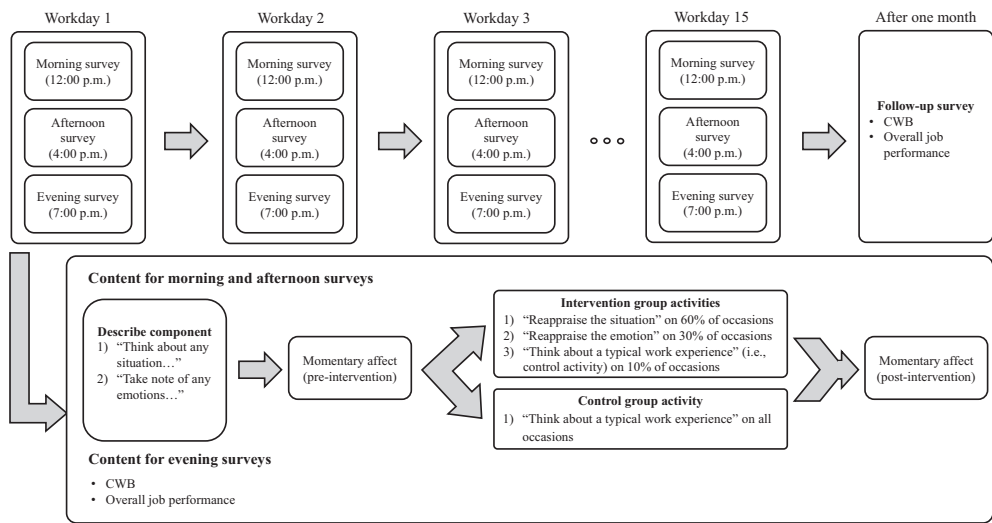


FIGURE 2 Study procedure. CWB, counterproductive work behaviour.

Procedure

We conducted a between-person randomized controlled trial to evaluate the effectiveness of an intervention package, following previous recommendations favouring such a design (Collins et al., 2005; Klasnja et al., 2015). Participants were randomly assigned to either the ecological momentary intervention group ($N=88$) or the control group ($N=88$). Within the between-person intervention group itself, we also employed within-person randomization of two cognitive reappraisal strategies and a control activity across time points (i.e. occasions) to compare the effectiveness of the two cognitive reappraisal strategies against each other and against the control activity. Both between-person groups underwent within-person measurement of focal variables during the intervention period and between-person measurement of performance outcomes 1 month after the intervention. Specifically, for both between-person groups, we sent an initial survey to all participants, which included the consent form, person-level measures for demographics, and instructions on how to complete daily surveys and how to respond to the intervention/control activity. On the following Monday, we emailed participants three surveys per day for three workweeks (i.e. 15 consecutive workdays, 45 total surveys). Each weekday, we distributed the surveys pertaining to the morning, afternoon and whole workday (hereafter, ‘morning survey’, ‘afternoon survey’ and ‘evening survey’) at 12 PM, 4 PM and 7 PM, respectively. This daily survey schedule allowed us to capture a representative sample of experiences and situations across the workday and aligned with common designs in experience sampling research (Gabriel et al., 2019). Moreover, using an experience sampling design (i.e. within-person measurement) in both between-person groups allowed us to capture the within-person fluctuations in negative and positive affect and their within-person relationships with performance outcomes. We sent a single follow-up survey (i.e. between-person measurement) 1 month after the conclusion of the intervention period. Figure 2 shows the study procedure.

We included interventions (or control activities) twice per day, in the morning and afternoon, to align with ‘natural breaks in participants’ days to facilitate recall’ in experience sampling studies (Fisher & To, 2012, p. 869). This permitted more than one opportunity to practice emotion regulation across each workday. For participants in the between-person intervention group, we provided three activities throughout the intervention period, which included (1) reappraising the situation, (2) reappraising the emotion and (3) a neutral control activity involving recalling a typical work experience (i.e. a within-intervention group control occasion). To clarify, participants in the

between-person intervention group completed one of the two intervention activities or the control activity on a given occasion—and, in total, completed both reappraisal activities on numerous occasions over the course of the study. We included a control activity on 10% of occasions *within* the between-person intervention group to (1) facilitate the exploratory analyses necessary to answer Research Question 1, which was tested within the intervention group only and (2) prevent participants in the intervention group from becoming habituated to the intervention activities (Nahum-Shani et al., 2018). In contrast, participants in the between-person control group always received the control activity (i.e. on 100% of occasions).

In the morning and afternoon surveys during the 3-week intervention period, participants in the between-person intervention and control groups underwent an identical procedure to reduce the demand effect (Weber & Cook, 1972). Specifically, all participants first described either an affective situation or an affective state they experienced in the past few hours (the initial description phase),² then reported their current NA and PA (pre-intervention/control affect), then received a cognitive reappraisal or control activity (the reappraisal phase), then reported their current NA and PA again (post-intervention/control affect), and finally reported their receptivity to the intervention or control activity.³

In the initial description phase, we chose to use a neutral prompt to ask participants to describe any events or affective states (e.g. positive, neutral, negative events and affect), as opposed to solely negative events or affect, to avoid measurement reactivity, which refers to the potential for participants' responses to be influenced by the measurement process itself (French & Sutton, 2010), for all participants before the delivery of a cognitive reappraisal or control activity. In the reappraisal phase, for occasions involving reappraising the situation intervention within the between-person intervention group, we instructed participants to think about any positive or pleasant aspects or outcomes of the situation they described in the previous phase. Because of the general emphasis on the positive aspects of the situations in the instruction, reappraising the situation can make (1) negative, stressful situations less negative and (2) neutral or positive situations more positive. In contrast, for occasions involving reappraising the emotion intervention within the between-person intervention group, we asked participants to openly accept any negative affective experiences they described, without any judgement. Our wording of the instructions for these two reappraisal activities was adapted from existing reappraisal research (Webb et al., 2012). For the control activity (experienced on 10% of occasions in the between-person intervention group and 100% of occasions in the between-person control group), following best-practice recommendations (see Quoidbach et al., 2015), we adopted an 'active' yet neutrally valenced activity, wherein participants were asked to recall a typical work experience in the past few hours, rather than simply an absence of any intervention.

For the between-person intervention group, we randomized these three activities—reappraising the situation, reappraising the emotion and the control activity—on a given occasion, with occurrence probabilities of 60%, 30% and 10%, respectively. We chose to deliver the reappraising the situation activity more frequently (within-person) than the reappraising the emotion activity, given the particular relevance of events to workplace affect (Weiss & Cropanzano, 1996) and to more intense affective episodes (Russell, 2003; Weiss & Merlo, 2020). To maximize the overall effectiveness of the intervention, we only included the control activity on 10% of occasions within the

²Of note, we purposefully asked participants to reappraise *any* situations or emotions from the past few hours rather than specifically work-related situations or affect (see Table 1) because both work events (e.g. being yelled at by one's supervisor) and non-work events (e.g. receiving a phone call from the daycare centre and thereby learning that one's child is sick) can influence workplace affect and, in turn, performance. Thus, we wanted to ensure that our reappraisal instructions referred to any affective events that participants *experienced* during their work hours (Weiss & Cropanzano, 1996).

³The pre- and post-intervention affect were assessed in the same survey, together with the intervention/control activities. Within each survey, the sequence followed a clear temporal order: participants reported their pre-intervention affect, engaged with the intervention or control activity, and then reported their post-intervention affect. We strictly maintained this temporal order by presenting the questions and activities on different pages of the online survey and not allowing participants to go back to a previous page. We chose not to send additional surveys with a time lag to reduce participant burden and maintain a high response rate. We believe that this approach balances methodological rigor with participant engagement.

TABLE 1 Emotion regulation intervention activities.

Intervention activity	Initial description	Cognitive reappraisal activity (first two sections of table) or control activity (third section of table)
Reappraise the situation	Please take a minute or two to think about any situation from the last few hours and imagine that you are in the situation again. Once you have thought about this situation, please briefly describe it in a sentence or two. Please note that you will not be able to advance to the next page until you have written a minimum of five words	Now, please think about any positive or pleasant aspects or outcomes of the situation you described. It is not uncommon for negative situations to have some positive outcomes. So, even if the situation was negative or unpleasant, is there anything you found out that is good for you, anything you learned from the situation, or anything you feel you would rather have or experience now than later? We encourage you to think about at least two positive aspects of the situation (e.g. that you did not notice before), but please do not worry if you are not able to think of that many positive aspects. After reflecting on the positive aspects/outcomes of the situation for a minute or two, please briefly describe those positive aspects in one or two sentences. As an example, you might respond as follows: ‘This morning, after I spilled coffee on my keyboard, causing it to stop working, our computer people were able to get me a new keyboard surprisingly quickly—and, while waiting, I got to take a short break from work. The incident was also useful because it finally helped me switch to using a coffee mug that has a lid’. Please note that you will not be able to advance to the next page until you have written a minimum of 15 words
Reappraise the emotional response	Please take note of any emotions from the last few hours. Once you have reflected on your feelings, please briefly describe them in a sentence or two—with a focus on the emotions themselves (e.g. joy, contentment, anger, sadness, fear, disgust) rather than their cause. Please note that you will not be able to advance to the next page until you have written a minimum of five words	Now, try to openly accept any negative emotions you described rather than judging whether or not they are ‘bad’ or ‘unwanted’. Remember that emotions are normal, healthy and temporary reactions that come and go. After reflecting on your emotions for a minute or two, please briefly describe them in a non-judgmental and accepting way in one or two sentences. As an example, you might respond as follows: ‘I felt a few different negative emotions over the last few hours. I felt frustrated this morning, but that’s okay: frustration happens and is normal. At other times, I felt a little tired and sad, but I know emotions like this come and go’. Please note that you will not be able to advance to the next page until you have written a minimum of 15 words
Control	One of the two descriptions above, with an equal probability (i.e. 50% and 50%)	Please take a minute or two to think about a typical experience you’ve encountered from the last few hours. Please briefly describe your experience in a sentence or two. Please note that you will not be able to advance to the next page until you have written a minimum of 15 words

Note: For the between-person intervention group, we randomized these three intervention activities at a given time point with probabilities of 60%, 30% and 10%, respectively. This was done to (1) facilitate the exploratory analyses, *within* the between-person intervention group only, comparing the two reappraisal interventions to each other and the control activity and (2) prevent participants in the between-person intervention group from becoming overly bored with or habituated to the intervention activities (Nahum-Shani et al., 2018). Individuals in the control group were exposed to the control activity on 100% of occasions.

between-person intervention group. Conversely, as indicated previously, individuals in the control group were exposed to the control activity on 100% of occasions. Table 1 presents the verbatim instructions for these activities.

In the evening surveys, we measured daily CWB and daily overall job performance. To assess the possibility of ‘fade out’ of intervention effects, we also measured CWB and overall job performance in a one-time follow-up survey 1 month after the intervention period ended. The 1-month interval was determined based on (1) the time interval used in another cognitive reappraisal ecological momentary intervention on well-being outcomes (Kam et al., 2024) and (2) the moderate within-person variability in job performance shown in the literature (Dalal et al., 2020).

We aggregated the observation-level pre- and post-intervention NA and PA scores across morning and afternoon surveys in a given day to form a composite at the day level ($N = 2100$ daily observations), which we then paired with that day's evening survey (which already had a day-level timescale; $N = 1836$ observations). Therefore, the total sample size for analyses, including performance outcomes at the day level, was 1836 daily observations (response rate = 70%).

Measures

Negative and positive affect

We used the Scale of Positive and Negative Experience (SPANE; Diener et al., 2009) to measure pre- and post-intervention affect in the morning and afternoon surveys. Participants were asked to report how much they were experiencing each of 12 feelings at the moment. Sample items include ‘positive’, ‘negative’, ‘pleasant’ and ‘unpleasant’.

Counterproductive work Behaviour

We used Dalal et al.'s (2009) measure of CWB in the evening surveys and the follow-up survey. Participants were asked to indicate whether or not they performed each of 12 forms of work behaviour ‘today’ (in the evening surveys) or over the past 1 month since the interventions ended (in the follow-up survey) on a dichotomous scale (0 = *no* and 1 = *yes*). A sample item was ‘I behaved in an unpleasant manner toward another employee’.

Overall job performance

We used Dalal et al.'s (2009) single-item measure of overall job performance in the evening surveys (‘today’) and the follow-up survey (‘over the past month’). Participants were asked to rate their overall job performance on a 1 (*very low*) to 5 (*very high*) scale. The item was ‘My overall job performance was....’

Receptivity

We developed one item for each variable involving receptivity to the intervention activity. Specifically, the item for availability was ‘In thinking about your workload, how much time did you have to perform that emotion regulation activity?’ Responses were collected on a 5-point Likert scale, from 1 (*none at all*) to 5 (*a lot*). The item for the perceived usefulness of the intervention was ‘You just completed an emotion regulation activity. Before completing this activity, how beneficial would you have thought the emotion regulation activity would have been?’ Responses are on a 5-point Likert scale, ranging from 1 (*not at all likely*) to 5 (*extremely likely*). Receptivity was measured in the morning and afternoon surveys.

Analysis plan

The nested structure of the data (i.e. daily observations at Level 1 nested in persons at Level 2) necessitated a multilevel approach (Preacher et al., 2010). To test Hypotheses 1–5, we conducted multilevel path analysis and did not employ person-mean centring, as the partitioning of within- and between-person variance is done implicitly in the modelling procedures (Preacher et al., 2010). We estimated a 2-1-1 mediation model for short-term (daily) performance outcomes and a 2-1-2 mediation model for long-term (1-month post-study) performance outcomes. To test the effects of the intervention, we created a dichotomous variable coded such that 0 = between-person control group and 1 = between-person intervention group. This variable served as the independent variable in both mediation models. In both models, we specified the direct effects of the intervention on post-intervention NA and PA (controlling for pre-intervention NA and PA, respectively) and the indirect effects of the intervention on CWB and overall performance through post-intervention NA and PA. We also specified the direct effects of the intervention on short- and long-term performance outcomes, respectively, for the sake of completeness. The significance of indirect effects was tested using 95% confidence intervals calculated via the Monte Carlo method (Zhang et al., 2009). The fit of the short-term model was as follows: $\chi^2(14) = 371.88$, RMSEA = .12, SRMR_{within} = .04, SRMR_{between} = .02, AIC = 5693.46, BIC = 5858.92, CFI = .94, TLI = .86. The fit of the long-term model was as follows: $\chi^2(9) = 100.56$, RMSEA = .07, SRMR_{within} = .02, SRMR_{between} = .02, AIC = 5787.59, BIC = 5922.83, CFI = .99, TLI = .96.

To test Hypothesis 6 and Research Question 1, we conducted multilevel modelling (MLM) with person-mean centred within-person predictors, their person means and grand-mean centred between-person predictors (Enders & Tofighi, 2007). To test Hypothesis 6 on the moderating role of receptivity, we utilized MLM to compute one model for NA and one model for PA, using data in both between-person intervention and control groups. We created two interaction terms by multiplying each person-mean-centred receptivity indicator (i.e. availability and perceived usefulness) with the grand-mean-centred intervention variable. Additionally, we included the direct effects of the two receptivity variables on post-intervention NA and PA and controlled for pre-intervention NA and PA (via person-mean centred terms and person means). Of note, both moderators were tested simultaneously in each model.

To test Research Question 1 on the effectiveness of the two cognitive reappraisal strategies—both against each other and against the control—we estimated a multi-level model, with Level 1 representing not the day level but rather the within-day observation level (i.e. before and after the intervention in the mornings and afternoons) and Level 2 representing the person level. For this analysis, we only included participants in the between-person intervention group ($N = 88$), who had themselves received the two types of cognitive reappraisal activities and the control activity on different occasions. Two dummy-coded predictors were used, such that 1 = a reappraisal activity and 0 = the other reappraisal activity or the control activity. All analyses were conducted in R (R Core Team, 2023) using the *lavaan* package (Rosseel, 2012) and the *lme4* package (Bates et al., 2015). Data supporting the results are provided on the Open Science Framework at https://osf.io/eyh9x/?view_only=6acc660d833646988fd4bd4e0c89f631.

RESULTS

We calculated descriptive statistics for all variables as well as intra-class correlations (ICCs) for all within-person variables (see Table 2). The ICC(1) values ranged from .33 to .68, indicating that significant proportions of variance in all the within-person variables are attributable to both within- and between-person sources (LeBreton & Senter, 2008), and therefore that a multilevel framework should be adopted. We also calculated within- and between-person correlations (see Table 3).

TABLE 2 Descriptive statistics for study variables.

Variable	Full sample				Intervention		Control		
	<i>M</i>	<i>SD</i>	α	ICC	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
Between-person intervention group	0.50	0.50	–	–					
Pre-intervention NA	1.30	0.41	0.86 0.95	0.58	1.31	0.42	1.30	0.40	
Pre-intervention PA	2.29	0.84	0.94 0.97	0.68	2.26	0.78	2.32	0.90	
Post-intervention NA	1.27	0.40	0.87 0.94	0.58	1.26	0.40	1.29	0.41	
Post-intervention PA	2.36	0.80	0.95 0.97	0.66	2.36	0.73	2.37	0.88	
CWB (short term)	0.12	0.14	–	0.64					
Overall job performance (short term)	3.36	0.53	–	0.33					
CWB (long term)	0.23	0.27	–	–					
Overall job performance (long term)	3.57	0.72	–	–					
Availability	2.83	0.84	–	0.61					
Perceived usefulness	2.56	0.88	–	0.69					

Note: Within-person N (Level 1) = 1836–2112. Between-person N (Level 2) = 171–176. Between-person intervention group variable is dummy coded such that 0 = Between-person control group and 1 = Between-person intervention group. In places where there are two alpha (i.e., α) values, the value to the left of the vertical bar is the within-person alpha, and the value to the right of the bar is the between-person alpha (Geldhof et al., 2014). The short-term outcome model includes CWB and overall job performance measured at the daily level, whereas the long-term outcome model includes CWB and overall job performance measured 1 month after the conclusion of the intervention program (so as to assess the possibility that the intervention effect ‘fades out’ soon after the interventions cease).

Abbreviations: CWB, counterproductive work behaviour; ICC, intraclass correlation; *M*, mean; NA, negative affect; PA, positive affect; *SD*, standard deviation.

The effects of the cognitive reappraisal intervention

Hypotheses 1a,b predicted that the cognitive reappraisal intervention would decrease post-intervention NA and increase post-intervention PA, respectively. As can be seen in Table 4, the intervention significantly decreased post-intervention NA and significantly increased post-intervention PA, supporting Hypothesis 1a,b, respectively.

Hypotheses 2a,b predicted that post-intervention NA would be positively related to CWB and negatively related to overall job performance, respectively. Hypothesis 3 predicted that post-intervention PA would be positively related to overall job performance.⁴ As reported in Table 4, post-intervention NA was positively related to short-term and long-term CWB while demonstrating no relationship with short-term or long-term overall job performance. Post-intervention PA was positively related to short-term and long-term overall job performance. Therefore, Hypotheses 2a and 3 were supported, whereas Hypothesis 2b was not supported.

Hypotheses 4a,b predicted that post-intervention NA would mediate the relationships of the cognitive reappraisal intervention with CWB and overall job performance, respectively. Hypothesis 5 predicted that post-intervention PA would mediate the relationship between the cognitive reappraisal intervention and overall job performance.⁵ As reported in Table 5, the indirect effects of the intervention, through post-intervention NA, were significant and negative for short-term and long-term CWB but not significant for short-term or long-term overall job performance. Furthermore, the indirect

⁴In exploratory fashion, we did also include a path from post-intervention PA to CWB in the model. As can be seen in the results presented in Table 4, consistent with previous theorizing (e.g., Spector & Fox, 2005) and empirical tests (e.g., Dalal et al., 2009; Sawhney & McCord, 2023), there was no significant relationship between post-intervention PA and (short-term or long-term) CWB.

⁵As suggested by a reviewer, we also tested an alternative model that additionally controlled for pre-intervention PA and pre-intervention NA when testing the intervention's effect on post-intervention NA and post-intervention PA, respectively (i.e. additionally controlling for the *opposite* pre-intervention affective state). Substantive results pertaining to direct and indirect effects did not change, and thus we report the more parsimonious model in this paper.

TABLE 3 Correlations for study variables.

Variable	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
1. Between-person intervention group	–	.03	–.05	–.02	–.01	–.03	.04	–.01	.12	.01	–.04
2. Pre-intervention NA	–	–	–.06	.98***	–.09	.62***	–.10	.40***	–.03	–.03	.01
3. Pre-intervention PA	–	–.43***	–	–.01	.98***	–.02	.60***	–.02	.42***	.21	.30
4. Post-intervention NA	–	.82***	–.36***	–	–.06	.62***	–.07	.40***	.01	–.03	.03
5. Post-intervention PA	–	–.40***	.85***	–.44***	–	–.07	.59***	–.08	.37***	.24	.35
6. CWB (short term)	–	.19***	–.10***	.19***	–.13***	–	–.25***	.76***	–.08	–.04	–.24
7. Overall job performance (short term)	–	–.17***	.18***	–.15***	.18***	–.28***	–	–.20**	.64***	.19	.39
8. CWB (long term)	–	–	–	–	–	–	–	–	.01	–.11	–.32
9. Overall job performance (long term)	–	–	–	–	–	–	–	–	–	.09	.06
10. Availability	–	–.06	.05	–.08	.07	.02	–.10	–	–	–	.44
11. Perceived usefulness	–	–.10	.08	–.11	.16	.01	.12	–	–	–	–

Note: Within-person N (Level 1) = 1836–2112. Between-person N (Level 2) = 171–176. Between-person correlations are above the diagonal, within-person correlations are below the diagonal. The intervention was a between-person manipulation, and long-term CWB and overall job performance were measured at the between-person level; hence, within-person correlations do not exist for these variables. The between-person intervention group variable is dummy coded such that 0 = between-person control group and 1 = between-person intervention group. The short-term outcome model includes CWB and overall job performance measured at the daily level, whereas the long-term outcome model includes CWB and overall job performance measured 1 month after the end of the intervention program (so as to assess the possibility that the intervention effect ‘fades out’ soon after the interventions cease).

Abbreviations: CWB, counterproductive work behaviour; NA, negative affect; PA, positive affect.
* $p < .05$. ** $p < .01$. *** $p < .001$.

TABLE 4 Results of multilevel path analysis for short-term and long-term performance outcomes.

Variable	Post-intervention NA		Post-intervention PA		CWB		Overall job performance	
	γ	SE	γ	SE	γ	SE	γ	SE
Short-term outcome model								
Intercept	0.04	(0.03)	0.18***	(0.04)	−0.17***	(0.04)	2.54***	(0.16)
Level 2								
Pre-intervention NA	0.97***	(0.02)	—	—	—	—	—	—
Pre-intervention PA	—	—	0.94***	(0.02)	—	—	—	—
Between-person intervention group	−0.04**	(0.02)	0.06*	(0.03)	0.01	(0.02)	0.04	(0.07)
Post-intervention NA	—	—	—	—	0.23***	(0.02)	−0.09	(0.09)
Post-intervention PA	—	—	—	—	−0.00	(0.01)	0.39***	(0.04)
Level 1								
Pre-intervention NA	0.78***	(0.01)	—	—	0.05***	(0.01)	−0.20**	(0.06)
Pre-intervention PA	—	—	0.85***	(0.01)	−0.01*	(0.01)	0.18***	(0.03)
Long-term outcome model								
Intercept	0.04+	(0.02)	0.15***	(0.04)	−0.16+	(0.09)	2.51***	(0.24)
Level 2								
Pre-intervention NA	0.97***	(0.02)	—	—	—	—	—	—
Pre-intervention PA	—	—	0.95***	(0.01)	—	—	—	—
Between-person intervention group	−0.05***	(0.01)	0.06*	(0.02)	0.02	(0.04)	0.19+	(0.10)
Post-intervention NA	—	—	—	—	0.29***	(0.05)	0.11	(0.13)
Post-intervention PA	—	—	—	—	0.01	(0.02)	0.36***	(0.06)
Level 1								
Pre-intervention NA	0.78***	(0.01)	—	—	—	—	—	—
Pre-intervention PA	—	—	0.84***	(0.01)	—	—	—	—

Note: Within-person level N (Level 1) = 1836 for proximal outcomes (NA and PA) and 2070 for distal outcomes (CWB and overall job performance). Between-person level N (Level 2) = 175 for proximal outcomes (NA and PA) and 172 for distal outcomes (CWB and overall job performance). Between-person intervention group variable is dummy coded such that 0 = between-person control group and 1 = between-person intervention group. Pre-intervention NA and PA are included as control variables in models featuring post-intervention NA and PA, respectively. The short-term outcome model includes CWB and overall job performance measured at the daily level, whereas the long-term outcome model includes CWB and overall job performance measured 1 month after the end of the intervention program; post-intervention NA and PA were the same in both short-term and long-term models (i.e. the same measurement timepoint).

Abbreviations: CWB, counterproductive work behaviour; NA, negative affect; PA, positive affect.

* $p < .10$. ** $p < .05$. *** $p < .001$.

effects of the intervention, through post-intervention PA, were significant and positive for short-term and long-term overall job performance. Therefore, Hypotheses 4a and 5 were supported, whereas Hypothesis 4b was not supported.^{6,7}

⁶As suggested by a reviewer, we also tested an alternative model that excluded data from days when a person assigned to the between-person intervention group received a control activity at either the morning or the afternoon survey (or both). In other words, in this alternative model, the between-person intervention group received cognitive reappraisal interventions on 100% (rather than the original 90%) of occasions. Substantive conclusions remained identical, suggesting that our findings are robust to the inclusion of the control activity on 10% of occasions within the between-person intervention group (which was necessary to permit a test of Research Question 1).

⁷As an additional sensitivity check, we reran focal mediation models excluding days on which participants wrote fewer than one standard deviation ($SD = 222$ words) below the average number of word counts ($M = 393$ words) in response to the reappraisal prompts (regardless of whether they were reappraising the situation, reappraising the emotion, or performing a control activity). Note that this average number of words includes responses to both morning and afternoon surveys—that is, the mean and standard deviation are day-level rather than occasion-level scores. In this analysis, the Level 1 N drops from 2100 to 1957 daily observations. Substantive conclusions remained the same. This supports the quality of the data with respect to engagement in the intervention.

TABLE 5 Results of multilevel path analysis for indirect (mediated) effects.

Mediated effect	Indirect effect	
	Estimate	95% CI
Short-term outcomes		
Intervention → Post-intervention NA → CWB	−0.010	[−0.017, −0.003]
Intervention → Post-intervention NA → Overall job performance	0.004	[−0.004, 0.014]
Intervention → Post-intervention PA → CWB	0.000	[−0.001, 0.001]
Intervention → Post-intervention PA → Overall job performance	0.021	[0.002, 0.041]
Long-term outcomes		
Intervention → Post-intervention NA → CWB	−0.013	[−0.022, −0.005]
Intervention → Post-intervention NA → Overall job performance	−0.005	[−0.018, 0.007]
Intervention → Post-intervention PA → CWB	0.000	[−0.003, 0.004]
Intervention → Post-intervention PA → Overall job performance	0.020	[0.003, 0.039]

Note: Intervention variable is dummy coded such that 0 = Between-person control group and 1 = Between-person intervention group. The short-term outcome model includes CWB and overall job performance measured at the daily level, whereas the long-term outcome model includes CWB and overall job performance measured 1 month after the conclusion of the intervention program.

Abbreviations: CWB, counterproductive work behaviour; NA, negative affect; PA, positive affect.

The moderating effect of receptivity

To test Hypotheses 6a,b, we examined whether an employee's level of availability for the intervention and perceived usefulness of the intervention (pertaining to the ability and motivation aspects of receptivity towards the intervention, respectively) moderated the effect of the intervention on post-intervention NA and PA, after controlling for pre-intervention NA and PA, respectively. Results are reported in Table 6. We found no significant moderating effect of availability on the impact of the intervention on post-intervention NA or PA. There was also no direct effect of availability on post-intervention NA or PA.

Similarly, there was no significant moderating effect of perceived usefulness on the impact of the intervention on post-intervention NA or PA. There was also no direct effect of perceived usefulness on post-intervention NA. However, there was a significant direct effect of perceived usefulness on post-intervention PA, suggesting that higher perceived usefulness of the activity, whether intervention or control, was associated with higher post-intervention PA. In summary, the results did not support the idea that availability or perceived usefulness of the intervention enhanced the effect of the intervention on post-intervention NA or PA. Thus, Hypotheses 6a,b were not supported.

Exploratory comparison of the two interventions

To test Research Question 1, we examined whether the intervention activities (i.e. reappraising the situation vs. reappraising the emotion) differed in predicting post-intervention NA and PA, after controlling for pre-intervention NA and PA. This test also allowed us to determine whether observed effects in the between-person intervention group were due to completing reappraisal activities or control activities, as the analysis utilized two dummy-coded variables with the control variable as the reference or baseline category. Specifically, one dummy variable was scored such that 1 = reappraising the situation and 0 = 'other' (i.e., reappraising the emotion or the control activity), whereas the other dummy variable was scored such that 1 = reappraising the emotion and 0 = 'other' (i.e. reappraising the situation or the

TABLE 6 Results of multilevel analysis for the moderating effect of receptivity.

Variable	Post-intervention NA		Post-intervention PA	
	γ	SE	γ	SE
Intercept	0.01	(.04)	−0.06	(0.05)
Level 2				
Pre-intervention NA	0.95***	(0.02)	—	—
Pre-intervention PA	—	—	0.94***	(0.01)
Between-person intervention group	−0.04**	(0.01)	0.06**	(0.02)
Availability	−0.01	(0.01)	0.02	(0.02)
Perceived usefulness	0.01	(0.01)	0.09***	(0.02)
Level 1				
Pre-intervention NA	0.73***	(0.03)	—	—
Pre-intervention PA	—	—	0.85***	(0.02)
Availability	−0.01	(0.01)	0.02	(0.01)
Perceived usefulness	−0.02	(0.01)	0.08***	(0.02)
Availability × between-person intervention group	0.01	(0.02)	0.03	(0.03)
Perceived usefulness × between-person intervention group	0.02	(0.02)	0.01	(0.04)

Note: Within-person level N (Level 1) = 2100; between-person level N (Level 2) = 176. Between-person intervention group variable is dummy coded such that 0 = between-person control group and 1 = between-person intervention group. Pre-intervention NA and PA are included as control variables in models featuring post-intervention NA and PA, respectively.
* $p < .05$. ** $p < .01$. *** $p < .001$.

TABLE 7 Results of multilevel analysis comparing the two intervention activities.

Variable	Post-intervention NA		Post-intervention PA	
	γ	SE	γ	SE
Intercept	0.37	(0.50)	−0.40	(0.76)
Level 2				
Pre-intervention NA	0.85***	(0.04)	—	—
Pre-intervention PA	—	—	0.95***	(0.03)
Intervention #1 (reappraising the situation)	1.11	(0.67)	0.25	(1.50)
Intervention #2 (reappraising the emotion)	−2.29**	(0.86)	1.36	(0.98)
Level 1				
Pre-intervention NA	0.73***	(0.01)	—	—
Pre-intervention PA	—	—	0.84***	(0.01)
Intervention #1 (reappraising the situation)	−0.05*	(0.02)	0.08*	(0.03)
Intervention #2 (reappraising the emotion)	−0.02	(0.02)	−0.02	(0.03)

Note: Level 1: N = 2005. Level 2: N = 88. This analysis was conducted at the decision-point level of analysis and only for participants in the between-person intervention group. Each intervention variable (Intervention #1 and Intervention #2) was dummy coded such that 1 = a reappraisal activity and 0 = the other reappraisal activity or the control activity.
Abbreviations: CWB, counterproductive work behaviour; NA, negative affect; PA, positive affect.
* $p < .05$. ** $p < .01$. *** $p < .001$.

control activity). As mentioned previously, for this analysis, Level 1 (the within-person level) was the observation level, not the day level. As reported in Table 7, reappraising the situation was found to be the more effective intervention activity, having significant within-person effects on post-intervention NA and PA. Reappraising the emotion, on the other hand, did not have a significant within-person

effect on post-intervention NA or PA (although it did have a significant between-person effect on post-intervention NA). This demonstrated that the primary effect of the intervention was driven specifically by reappraising the situation, rather than by reappraising the emotion or the control activity.

DISCUSSION

The current study examined the potential of a cognitive reappraisal ecological momentary intervention to provide timely support to mitigate NA and enhance PA in the workplace, and in turn mitigate CWB and enhance overall job performance. First, as hypothesized, the cognitive reappraisal intervention significantly reduced NA and increased PA. Second, as hypothesized, the cognitive reappraisal intervention exerted indirect effects on CWB via NA and on overall job performance via PA (but, contrary to expectations, did not exert indirect effects on overall job performance via NA). Third, contrary to expectations, receptivity was not found to strengthen the effect of the cognitive reappraisal intervention on reduced NA and increased PA. Fourth, an exploratory comparison of the two cognitive reappraisal strategies revealed that reappraising the situation was more effective than reappraising the emotion.

At a broader level, this study contributed beyond previous research by providing strong evidence of causality while being conducted *in situ* in the workplace. This approach allowed us to empirically test within-person affect theories using a multilevel research design that aligns with the appropriate levels of analysis and timeframes of the phenomenon (Kuppens et al., 2010; Weiss & Cropanzano, 1996).

Moreover, by measuring CWB and overall job performance at the end of each workday during the intervention period as well as 1 month after the intervention period, we captured a more comprehensive picture of the short- and long-term effects of ecological momentary interventions. We generally found similar short- and long-term effects, suggesting no 'fade out' of intervention effects over the 1 month after the end of the interventions. The absence of fade-out was possibly due to participants having developed self-regulatory skills during the intervention period and applying those skills so as to continue to successfully regulate their affect and behaviour after the end of the intervention period.

Regarding practical implications, a major advantage of the present intervention is its concise and accessible nature to participants. For example, such an ecological momentary intervention can be completed quickly and offered through a mobile device (Schenkel et al., 2022), making it a feasible tool in the workplace. Moreover, this study demonstrates the potential effectiveness of ecological momentary interventions in shaping employees' affective states in a near-real-time work context. In conclusion, our study supports that ecological momentary interventions are potential practical tools in organizational practice due to their feasibility and preliminary support for effectiveness.

Limitations and future directions

Despite the contributions of our study, it is necessary to acknowledge some limitations that offer fruitful avenues for future research. First, given that this study represents an initial exploration of ecological momentary interventions in organizational research, we did not tailor the delivery of momentary interventions based on employee receptivity on particular occasions. Instead, we measured receptivity after employees completed the intervention and tested its role as a moderator of intervention effectiveness, finding no significant effect. A *post-hoc* explanation for this unsupported hypothesis may be a restriction in the range of receptivity, as completing the intervention (vs. skipping the survey, accompanying intervention, and questions about receptivity) may imply a minimum level of availability for and perceived usefulness of the intervention. Future research can build on the current study by measuring receptivity *before* the delivery of interventions and, in fact, by delivering interventions only when participants are receptive. It is, however,

important to note that a downside to tailoring the delivery of interventions in such a manner is the introduction of treatment selection bias and, consequently, a decrease in internal validity (Zhu et al., 2024). Another downside to such tailoring involves the need to identify an appropriate cut score for receptivity on the basis of theory, extensive pilot testing and/or prior practice in the health behaviour literature.

Second, we acknowledge that some of the affective events reported in the cognitive reappraisal intervention may have ended prior to the intervention. Future research could therefore prompt participants to report events or affect experienced within a shorter timeframe, such as the past hour. However, such designs would also have their drawbacks relative to the current one: researchers would either have to send surveys much more often (e.g. every hour), which would be intrusive to participants and probably yield missing and poor quality data, or else send surveys covering only part of the workday, which would raise questions about coverage choices and potentially limit the between-person effectiveness of the intervention. We believe our current design represents a reasonable trade-off between participant engagement and intervention effectiveness. Future research could, however, explore novel approaches, such as passive data collection methods (e.g. wearable devices), to capture key moments while minimizing participant burden.

Third, our study included only cognitive reappraisal (albeit of two varieties) as an emotion regulation strategy. Although previous research has generally shown that cognitive reappraisal is among the most effective emotion regulation strategies (Webb et al., 2012), it is possible that other emotion regulation strategies may be more effective for certain workplace events and affective states—or that employees may be more receptive to other strategies. Future research can therefore broaden the scope of interventions to encompass other emotion regulation strategies, such as situation selection and modification (after pilot testing to examine the extent to which they are actually possible in the workplace), and deliver the right type of emotion regulation strategy based on an employee's transient affective needs and experienced situations.

Fourth, we did not assess the psychological characteristics of the events or types of stressors, limiting our ability to examine whether the cognitive reappraisal intervention functions differently in various situations. To enhance the specificity and effectiveness of the intervention, future research should draw on event-based theories and literatures (Morgeson et al., 2015; Parrigon et al., 2017; Reindl et al., 2021) and ask participants to evaluate key features of the situations they encounter, such as valence, importance, typicality and adversity. Similarly, future research can draw on the literature on daily work stressors (e.g. Almeida, 2005; Barling & Kryl, 1990) and customize the intervention support based on specific types of stressors. Here, again, it should be noted that although such within-person customization increases the precision and relevance of the intervention, it also introduces treatment selection bias and, consequently, challenges to internal validity (Zhu et al., 2024).

Fifth, as an initial study on this topic, we focused on broad measures of NA and PA, as opposed to discrete emotions. This approach, while informative, may overlook the nuanced role that specific emotions play in shaping responses to interventions. Future research could employ more fine-grained assessments to capture discrete emotions (e.g. anxiety, frustration, joy) and either develop cognitive reappraisal interventions tailored to each emotion or evaluate whether existing cognitive reappraisal interventions have differential effects on various emotions.

Sixth, although we followed best-practice recommendations to design the intervention and control activities (e.g. Quoidbach et al., 2015), alternative aspects of the intervention's design may have influenced these findings. In the first place, as noted by a reviewer, the reappraisal instructions were of different lengths across conditions: 195 words for reappraising the situation, 111 words for reappraising the emotion and 29 words for the control activity. It is possible that the differing lengths of the instructions influenced the perceived salience and hence effectiveness of the instructions. Future research should aim to unconfound the content and length of the intervention instructions.

Furthermore, the randomization probability of the intervention conditions meant that participants more frequently received the reappraising the situation activity than the reappraising the emotion activity. This was done in part due to the particular importance of events to workplace affect generally

(Russell, 2003; Weiss & Merlo, 2020), but a different randomization probability may have produced different results. Moreover, following prior cognitive reappraisal research (e.g. Kam et al., 2024), we chose a 1-month interval for the follow-up measure of job performance to test the fade-out effect of the intervention. However, it is possible that intervention effects would diminish over a longer period of time. Future research should therefore explore these possibilities.

We also did not assess participants' emotional intelligence. If emotional intelligence matters here, it could matter in one of two ways. One possibility is that a baseline level of emotion regulation ability is a prerequisite for successful emotion regulation and that emotional intelligence confers such an ability. If so, these (and indeed any) emotion regulation interventions would be *less* beneficial to employees who score low on emotional intelligence. However, an alternative possibility holds that these interventions (which are explicitly designed to be short, quick and easy to understand and execute) are already familiar to employees who score high on emotional intelligence while being unfamiliar yet readily accessible to employees who score low on emotional intelligence. If so, these interventions would be *more* beneficial to employees scoring low on emotional intelligence. Future research should therefore examine the potential moderating role of emotional intelligence.

CONCLUSION

In conclusion, our study offers initial insights into the efficacy of an ecological momentary intervention on workplace affect and work behaviours. The results from our study present valuable causal evidence linking NA and PA to job performance. Our findings underscore the potential for integrating ecological momentary interventions with a randomized controlled trial design to advance within-person organizational theories.

AUTHOR CONTRIBUTIONS

Ze Zhu: Conceptualization; methodology; writing – original draft; writing – review and editing; investigation; supervision; project administration; visualization. **John A. Aitken:** Formal analysis; project administration; investigation; writing – original draft; writing – review and editing; data curation; methodology. **JeongJin Kim:** Data curation; writing – review and editing; project administration. **Julia I. Baines:** Data curation; project administration; writing – review and editing. **Seth A. Kaplan:** Conceptualization; methodology; investigation; writing – review and editing; supervision; funding acquisition; project administration. **Reeshad S. Dalal:** Conceptualization; methodology; investigation; funding acquisition; supervision; writing – review and editing; project administration. **Jordan Hassani:** Project administration; data curation.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

We have provided an active link with data supporting the results in the final accepted manuscript.

ORCID

Ze Zhu  <https://orcid.org/0000-0003-1076-3215>

John A. Aitken  <https://orcid.org/0000-0001-8727-5442>

JeongJin Kim  <https://orcid.org/0000-0002-0562-2488>
 Julia I. Baines  <https://orcid.org/0000-0003-1934-4973>
 Seth A. Kaplan  <https://orcid.org/0000-0002-5976-6815>
 Reeshad S. Dalal  <https://orcid.org/0000-0003-4040-4686>

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